Prayog — Requirements → Jira → GitHub → Testing Traceability Matrix

CS455 Deliverable 3 · Group 3 · 6 November 2026

1. How the matrix is built

The chain is the one in the sprint plan section 12: **requirement ID** (Integrated Requirements Specification) → **Jira issue** (story or task) → **GitHub** (pull request or commit) → **code** → **automated test** → **result on 6 November**. Every column is generated, not typed, by `src/make_trace.py`:

Column	Source
Requirement(s)	requirement IDs (`FR-…`, `NFR-…`) in the backlog item's description (`PRAYOG_backlog_jira_import.csv`)
Jira, Status	the Jira export (Jira key = plan key + 4)
Code	source files on `main` that cite the requirement ID in a comment or docstring (189 distinct IDs are cited in code)
GitHub	merged pull requests whose branch names the Jira key (`#n` = `Dishaswamy/Prayog`; `wr#n` = the working repository `baymaxisme/Prayog`, which used plan keys and is converted), plus PRs mapped by content (#11 Steward, #12 ledger, #19 access purpose, #21/#22 scheduler, #23–#25 CLI, #26 web app); otherwise commits whose message names the key
Automated tests	test modules that exercise the story, from reading the tests (`src/trace_map.py`)
Result	pass/fail counts of those modules in the 6 November junit files

1.1 The critical requirements, end to end

For the course's 8–10 critical requirements, the full chain. All results are from the final Linux run (`evidence/junit_backend_linux.xml`).

Requirement	Jira	PR	Test that proves it	Module (tests passing)	Result
NFR-GPU-001, FR-GPU-055: one live holder per GPU unit	PRAYOG-53	#5, #21	`test_passes_racing_for_the_last_free_unit_allocate_it_once`	`test_claim` (27) + `test_concurrency` (6)	pass
FR-GPU-054: all-or-nothing multi-GPU claim	PRAYOG-53	#5	`test_four_requested_with_only_three_lockable_holds_nothing`	`test_claim` (27)	pass
FR-GPU-013 to 016: rejection before queueing	PRAYOG-47	#6	`test_impossible_request_is_rejected_on_the_spot_and_stores_nothing`	`test_requests` (60) + `test_jobs_api` (29)	pass
FR-GPU-044 to 049, 185: score formulas and rounding	PRAYOG-51	#4	`test_derived_bounds_match_the_specification`	`test_scoring` (28)	pass
NFR-GPU-009, 060: starvation bounds	PRAYOG-105	`d3-final`	`test_research_job_waiting_90_minutes_longer_always_ranks_first`	`test_scoring` (28)	pass
FR-GPU-060, 061: idempotent dispatch	PRAYOG-59	#9	`test_acknowledging_starts_the_run_once`	`test_dispatch` (20)	pass
FR-GPU-065 to 069: cancel with two-phase release	PRAYOG-63	#9	`test_cancelling_a_running_job_frees_its_gpus_only_after_the_node_confirms`	`test_dispatch` (20) + `test_agent_api` (18)	pass
FR-GPU-186: owner-claim guard	PRAYOG-53	#5	`test_guest_claim_rolls_back_when_the_owner_has_an_eligible_job`	`test_claim` (27)	pass
FR-GPU-077, 135, 136: node failure, and an outage is not one	PRAYOG-64	`d3-final`	`test_an_api_outage_longer_than_the_timeout_marks_no_node_down`	`test_reaper` (17) + `test_liveness` (9)	pass

Requirement	Jira	PR	Test that proves it	Module (tests passing)	Result
NFR-GPU-030, A-18: another user's job	PRAYOG-108	`d3-final`	`test_another_users_job_cannot_be_read_edited_or_cancelled`	`test_security` (25)	pass

2. Coverage of the chain

Link in the chain	Items	Share
Backlog items (77 stories, 26 tasks; epics excluded)	103	—
Done in the Jira export	87	84%
Carry at least one requirement ID	83	81%
Requirement cited in code on `main`	50	49%
Linked to a merged pull request or a commit naming the key	61	59%
Mapped to automated tests in the repository	84	82%
… of which every mapped test passes on 6 Nov	67	
… with at least one test that fails on Windows only (`os.killpg`, symlinks, permission bits, path separators; all pass on Linux)	8 (PRAYOG-30, -44, -45, -60, -61, -71, -72, -94)	
No automated test, but other evidence (CI steps, drills, probes, D3 documents)	18	
No evidence at all	6 Done items: PRAYOG-21, -38, -40, -41, -43, -69	

3. Gaps

Done in Jira but without an automated test in the repository: PRAYOG-17, -18, -19, -20, -21, -22, -23, -24, -38, -40, -41, -43, -56, -57, -68, -69, -74, -112 (18 items).

Most of these are platform tasks whose evidence is the CI workflow or a document rather than a test (PRAYOG-17…24). The functional gaps are PRAYOG-38 (access extension), -40 (one person holding admin and submitting rights), -41 (account deactivation), -43 (professor continuing access), -56 (show remaining cap), -57 (edit a queued job), -68 (tell the owner no node can ever run the job), -69 (CLI notifications) and -74 (node-down alert, exercised only by the live drills). They are the first items of the Sprint 3 test backlog (Project report section 6).

Steward stories (PRAYOG-95…102) are covered by the 222-test Steward suite, which is kept outside `main` at the team lead's request (branch `steward-full`), and by the 17-test backend–Steward harness; neither runs in CI on `main`.

Requirement coverage. The Integrated Specification defines 400 requirement IDs (SADD Appendix A traces all of them to design). On `main`, 189 are cited in code and 35 in tests; most tests are named by behaviour rather than by requirement ID, which is why the story-level matrix below maps tests by reading them.

4. D3 defects → fix → regression test

Defect	Requirement	Jira (after import)	Fix	Regression test	Result
D3-01	FR-STW-001, FR-STW-006, FR-INT-01	Bug → PRAYOG-14	`8a38f79` (PR #26)	`test_cli_rebuild_reaches_steward` (2)	pass
D3-02	FR-AUTH-60, FR-AUD-01	Bug → PRAYOG-6	`717f1fc` (PR #26)	`test_review_fixes` lockout (4), concurrency C4, probe A19	pass
D3-03	FR-GPU-055, FR-GPU-056	Bug → PRAYOG-8	`f358e01`	concurrency C6	pass (fails on `main`)
D3-04	FR-GPU-154	Bug → PRAYOG-9	`f358e01`	`test_job_list_batched` (3)	pass
D3-05	NFR-SEC (node isolation)	Bug → PRAYOG-9	`f358e01`	`test_node_agent_units` (POSIX)	pass in CI
D3-06	availability	Bug → PRAYOG-5	`f358e01`	`test_database_unavailable` (2), drill F3	pass
D3-07	performance	Bug → PRAYOG-5	`f358e01`	load re-run	0% errors
D3-08	FR-GPU-077	Bug → PRAYOG-11	`d3-final`	`test_liveness` (9), failure experiment	pass
D3-09	handover (PRAYOG-88)	Bug → PRAYOG-12	`717f1fc`	`test_review_fixes`	pass
D3-10	S1 approval (PRAYOG-98)	Bug → PRAYOG-14	`717f1fc`	`test_review_fixes`	pass
D3-11	FR-GPU-145 / active access	Bug → PRAYOG-6	`717f1fc`	`test_review_fixes` (2)	pass
D3-12	CI	Bug → PRAYOG-5	`cc209d9`	CI run on PR #26	green
D3-13	NFR-GPU-030	Bug → PRAYOG-9	`d3-final`	`test_security`, probe A03b	pass
D3-14	NFR-GPU-014	Bug → PRAYOG-8	`d3-final`	`test_adversarial` (3), `pass_benchmark`	pass
D3-15	NFR-GPU-039	Bug → PRAYOG-8	`d3-final`	`test_adversarial` (2), `load_test`	pass
D3-16	input validation	Bug → PRAYOG-15	`d3-final`	`test_adversarial` (4)	pass
D3-17	input validation	Bug → PRAYOG-9	`d3-final`	`test_adversarial` (4)	pass
D3-18	FR-ML-003	Bug → PRAYOG-13	`d3-final`	`test_adversarial` (1)	pass
D3-19	FR-AUTH-28	Bug → PRAYOG-6	`d3-final`	`test_adversarial` (1)	pass
D3-20	A-18	Bug → PRAYOG-9	`d3-final`	`test_security` (1)	pass

5. Story-level matrix

In *Result*, a failure count is a Windows-only failure (see Coverage report section 4); `wr#` PRs are in `baymaxisme/Prayog`, `#` PRs in `Dishaswamy/Prayog`.

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-17	Create GitHub repository, branch protection and PR template	—	Done	—	—	—	—	CI workflow `.github/workflows/mlflow-validation.yml`; branch protection on GitHub

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-18	Configure Jira–GitHub integration and smart commits	—	Done	—	—	—	—	smart-commit keys in branch names (PRAYOG-79…92)
PRAYOG-19	Database schema and migrations for the core data model	—	Done	—	—	—	—	CI: `alembic upgrade head` on empty PostgreSQL + `alembic check`; repeated locally 6 Nov
PRAYOG-20	Configuration module with startup invariant validation	NFR-GPU-053	Done	`config.py`	9fe8fa5	`scheduler/test_config`	15 pass	config validated at startup (`load_config`)
PRAYOG-21	Structured logging and correlation IDs	NFR-GPU-049	Done	—	—	—	—	no test found
PRAYOG-22	Developer environment: Docker Compose with simulated node agents	NFR-GPU-033	Done	`__init__.py` `simulated.py`	—	—	—	docker-compose.yml (postgres, steward-runtime); simulated agents used in drills
PRAYOG-23	Metrics endpoint	NFR-GPU-051	Done	`metrics.py` `passes.py` `models.py` …	9fe8fa5	`scheduler/test_jobs_api`	29 pass	`/metrics` exercised manually
PRAYOG-24	Cloud deployment on the free tier, with a runbook	—	Done	—	—	—	—	not verified (Deployment section 9)
PRAYOG-25	Register a Prayog account with institute email and password	FR-AUTH-01	Done	—	—	`test_auth_routes`, `test_login`	23 pass	
PRAYOG-26	Submit an enrolment request with purpose, Professor and period	FR-AUTH-05	Done	—	#19	`test_access_request_purpose`	22 pass	
PRAYOG-27	See the status of my enrolment request	—	Done	—	—	`test_access_request_purpose`	22 pass	
PRAYOG-28	Professor reviews and decides enrolment requests naming them	FR-AUTH-10	Done	—	—	`test_access_request_purpose`	22 pass	
PRAYOG-29	Platform Admin approves or rejects enrolment	FR-AUTH-14	Done	—	—	`test_access_request_purpose`	22 pass	
PRAYOG-30	Authenticate the CLI with prayog login	FR-AUTH-20	Done	`cli.py`	#23, #24	`test_cli_login`, `cli/test_login_logout`	29 pass, 1 fail	

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-31	Record each approved user's lab and account type	FR-GPU-139	Done	—	#19	`test_access_request_purpose`, `test_lab_scheduling`	32 pass	
PRAYOG-32	Enrol a node agent with a one-time token and node-scoped credential	FR-GPU-110	Done	`domain.py` `nodes.py` `models.py`	—	`test_gpu_agent_api`	26 pass	drill F1 (re-start with stored credential)
PRAYOG-33	Authorize every protected operation server-side	FR-AUTH-28, NFR-GPU-029	Done	—	—	`test_review_fixes`, `test_auth_routes`, `cli/test_protected_commands`	42 pass	probes A01–A06, A11
PRAYOG-34	Log out and have the server session end	FR-AUTH-47	Done	—	—	`test_auth_routes`, `test_review_fixes`	24 pass	probes A12–A13
PRAYOG-35	Time-bounded access that expires on its approved end date	FR-AUTH-36	Done	—	—	`test_access_request_purpose`, `test_cli_login`	31 pass	
PRAYOG-36	Distinguish expired authorization from wrong credentials	NFR-AUTH-09	Done	—	—	`test_cli_login`	9 pass	
PRAYOG-37	Retrieval survives expiry; commitment does not	FR-GPU-145	Done	—	—	`test_review_fixes`	14 pass	
PRAYOG-38	Request an access extension	FR-AUTH-42	Done	—	—	—	—	no test found
PRAYOG-39	Revoke a user's authorization	FR-AUTH-39	Done	—	—	`test_access_request_purpose`, `test_auth_routes`	32 pass	
PRAYOG-40	Prevent one person holding admin and submitting rights	FR-GPU-173	Done	—	—	—	—	no test found
PRAYOG-41	Deactivate a departed user's account	FR-AUTH-66	Done	—	—	—	—	no test found
PRAYOG-42	Review authentication and authorization audit records	FR-AUD-01	Done	`auth.py`	—	`test_review_fixes`, `test_pool_picture`, `test_admin_cancel_job`	24 pass	
PRAYOG-43	Professor keeps continuing institutional access	—	Done	—	—	—	—	no test found

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-44	Link a local project folder to Prayog	FR-GPU-001	Done	—	#23, #24	`cli/test_folder`	19 pass, 1 fail	
PRAYOG-45	Capture an immutable, content-addressed project snapshot	FR-GPU-006	Done	—	#23, #24	`cli/test_folder`, `test_project_versions`, `test_staging`	67 pass, 1 fail	
PRAYOG-46	Submit a GPU execution request for a linked project	FR-GPU-009	Done	—	#6	`scheduler/test_requests`, `scheduler/test_jobs_api`, `test_gpu_run_submission`	113 pass	
PRAYOG-47	Declare GPU count and walltime, and validate them	FR-GPU-010	Done	`requests.py`	b3d9c46	`scheduler/test_requests`, `scheduler/test_jobs_api`, `test_gpu_run_submission`	113 pass	probe A16
PRAYOG-48	Answer the submission on the spot	FR-GPU-017	Done	`requests.py`	b3d9c46	`scheduler/test_jobs_api`, `test_gpu_run_submission`	53 pass	load test: POST /jobs p95 0.30 s
PRAYOG-49	Eligibility evaluator	FR-GPU-030	Done	—	#4	`scheduler/test_eligibility`, `test_pass_equivalence`, `test_node_sharing`	22 pass	
PRAYOG-50	Priority class resolver P0–P3	FR-GPU-036	Done	`classes.py`	af13dd1	`scheduler/test_classes`, `test_lab_scheduling`	19 pass	
PRAYOG-51	Fair-share scoring policy with deterministic ordering	FR-GPU-043	Done	`passes.py`	9739a3a	`scheduler/test_scoring`, `test_pass_equivalence`, `test_lab_scheduling`	40 pass	
PRAYOG-52	Scheduling pass orchestrator and its triggers	FR-GPU-121	Done	`steward_bridge.py` `passes.py` `run.py` …	#22	`scheduler/test_passes`, `test_pass_equivalence`, `test_rebuild_scheduling_path`	34 pass	drill F4 (event-driven pass without the tick)
PRAYOG-53	Atomic claim transaction with the owner-claim guard	NFR-GPU-001, FR-GPU-051	Done	`claim.py`	#5, #21	`scheduler/test_claim`, `scheduler/test_allocation_properties`, `test_lab_scheduling`, `test_concurrency`	44 pass	concurrency C3, C5, C6
PRAYOG-54	Show why a queued job is waiting, and until when	FR-GPU-021	Done	`eligibility.py` `requests.py` `jobs.py`	#26	`scheduler/test_waiting`, `scheduler/test_jobs_api`, `test_job_list_batched`, `test_web_app`	49 pass	
PRAYOG-55	Refuse a submission that would exceed the group's GPU-hour cap	FR-GPU-104	Done	`requests.py`	9fe8fa5	`scheduler/test_requests`, `scheduler/test_adversarial`, `test_admin_cancel_job`	87 pass	quota path only indirectly

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-56	Show the group's remaining cap and per-member consumption	FR-GPU-178	Done	`requests.py` `jobs.py` `research.js`	9fe8fa5	`scheduler/test_requests`, `scheduler/test_jobs_api`	89 pass	
PRAYOG-57	Edit a queued job without farming queue age	FR-GPU-070	Done	`requests.py`	9fe8fa5	`scheduler/test_requests`	60 pass	
PRAYOG-58	Resubmit a past run with changed resources	FR-GPU-133	Done	—	#25	`test_gpu_rebuild`	22 pass	
PRAYOG-59	Dispatch an allocation idempotently to the node agent	NFR-GPU-019, FR-GPU-058	Done	`dispatch.py`	#9	`scheduler/test_dispatch`, `scheduler/test_agent_api`, `test_gpu_agent_api`	64 pass	
PRAYOG-60	Node agent materialises the workspace and executes the run	FR-GPU-062	Done	—	fe073a0	`scheduler/test_agent_api`, `test_node_agent_units`, `test_node_agent_docker`	67 pass	CI hardened-container step
PRAYOG-61	Enforce walltime and stop safely	FR-GPU-073	Done	`reaper.py`	fe073a0	`scheduler/test_reaper`, `scheduler/test_agent_api`, `test_node_agent_runs`	49 pass	
PRAYOG-62	Check what my job is doing right now	FR-GPU-018	Done	—	#26	`scheduler/test_jobs_api`, `scheduler/test_agent_api`, `test_web_app`, `test_job_list_batched`	60 pass	load test GET /jobs/{id}
PRAYOG-63	Cancel a job in any state	FR-GPU-065	Done	`cancel.py` `models.py`	fe073a0	`scheduler/test_dispatch`, `scheduler/test_adversarial`, `scheduler/test_security`, `test_concurrency`, `test_review_fixes`	86 pass	concurrency C5
PRAYOG-64	Requeue a job interrupted by hardware without losing its place	FR-GPU-077	Done	`domain.py` `nodes.py` `reaper.py`	—	`scheduler/test_reaper`, `scheduler/test_liveness`, `test_gpu_run_finalize`, `test_review_fixes`	57 pass	drill F1
PRAYOG-65	Account GPU-hours actually consumed	FR-GPU-097	Done	—	#7	`scheduler/test_lifecycle`, `scheduler/test_lifecycle_edges`, `test_lab_scheduling`, `test_pool_picture`	45 pass	
PRAYOG-66	Return the outcome and logs of every terminal run	FR-GPU-091	Done	—	—	`test_gpu_run_finalize`	17 pass	
PRAYOG-67	Notify the owner of maintenance that will stop their job	FR-GPU-109	Done	`requests.py` `jobs.py`	—	`test_job_list_batched`	3 pass	planned maintenance stop in the job view

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-68	Tell the owner when no machine can ever run their queued job	FR-GPU-117	Done	`nodes.py` `reaper.py`	—	`scheduler/test_nodes`	31 pass	
PRAYOG-69	Notification records surfaced by the CLI	FR-GPU-176	Done	`maintenance.py` `nodes.py`	—	—	—	no test found
PRAYOG-70	Capture run outputs and store them server-side	NFR-GPU-022, FR-GPU-086	Done	—	—	`test_gpu_run_finalize`, `test_mlflow_artifacts`	19 pass	
PRAYOG-71	Synchronize results into the project folder automatically	FR-GPU-091	Done	—	#25	`cli/test_runs`	66 pass, 1 fail	
PRAYOG-72	Report exactly which files block a synchronization	FR-GPU-177	Done	—	#25	`cli/test_runs`	66 pass, 1 fail	
PRAYOG-73	Register GPU nodes and their units	FR-GPU-025	Done	`nodes.py`	fe073a0	`scheduler/test_nodes`, `scheduler/test_agent_api`, `test_gpu_agent_api`, `test_pool_picture`	79 pass	
PRAYOG-74	Alert when a node fails to come online or stops reporting	FR-GPU-174	Done	`domain.py` `nodes.py` `models.py`	fe073a0	`scheduler/test_nodes`, `scheduler/test_reaper`, `scheduler/test_liveness`	57 pass	drill F1/F4: NODE_DOWN alerts raised
PRAYOG-75	Schedule, narrow or cancel maintenance windows	FR-GPU-081	Done	`maintenance.py`	fe073a0	`scheduler/test_agent_api`, `scheduler/test_reaper`, `test_pool_picture`, `test_hardware_investigation`	48 pass	
PRAYOG-76	Take a node out of service for repair	FR-GPU-080	Done	`config.py` `domain.py` `nodes.py` …	fe073a0	`scheduler/test_nodes`, `scheduler/test_reaper`, `test_hardware_investigation`	57 pass	
PRAYOG-77	Return a repaired node to service	FR-GPU-169	Done	`nodes.py` `models.py`	fe073a0	`scheduler/test_nodes`, `test_hardware_investigation`	40 pass	
PRAYOG-78	Decommission a node permanently	FR-GPU-112	Done	`nodes.py`	fe073a0	`scheduler/test_nodes`, `scheduler/test_reaper`, `test_node_sharing`	53 pass	
PRAYOG-79	Change a node's donation mode and teaching-pool flag	FR-GPU-111	Done	`nodes.py`	#14	`test_node_sharing`	5 pass	
PRAYOG-80	Read-only operational picture of the pool	FR-GPU-156	Done	`overview.py` `pool.js`	#15, #26	`test_pool_picture`	4 pass	
PRAYOG-81	Cancel any user's job with a recorded reason	FR-GPU-065	Done	`cancel.py` `models.py`	#13	`test_admin_cancel_job`	6 pass	

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-82	Investigate a suspected hardware fault	FR-GPU-158	Done	`hardware_investigation.py` `config.py` `app.py` …	#16	`test_hardware_investigation`	9 pass	
PRAYOG-83	Capture metrics continuously and seal snapshots automatically	FR-ML-001	Done	—	#12, wr#32	`test_mlflow_lifecycle`, `test_mlflow_metric_batch`	30 pass	
PRAYOG-84	Outbox relay and idempotent ledger consumer	FR-ML-001	Done	—	#12, wr#33	`test_mlflow_outbox`, `test_mlflow_ledger_consumer`, `test_mlflow_outbox_worker`	18 pass	
PRAYOG-85	Snapshot state machine with INCOMPLETE	FR-ML-007	Done	—	#12, wr#41	`test_mlflow_lifecycle`, `test_mlflow_snapshot_storage_failures`	29 pass	
PRAYOG-86	Professor reads the ledger of students naming them	FR-ML-004	Done	`mlflow_professor_ledger.py` `professor.js`	#12, #26, wr#37	`test_mlflow_professor`, `test_mlflow_professor_api`, `test_mlflow_professor_ledger`, `test_mlflow_professor_postgres`, `test_mlflow_professor_redaction`	26 pass	
PRAYOG-87	Admin sees volume and availability, never content	FR-ML-005	Done	`app.py` `config.py`	#12, wr#29	`test_admin_mlflow`	2 pass	
PRAYOG-88	Hand over project IP to the supervising Professor	FR-ML-008	Done	`research.js`	#12, #26, wr#34	`test_mlflow_handover`, `test_review_fixes`	19 pass	
PRAYOG-89	Trigger an environment rebuild from a Snapshot ID	FR-ML-003	Done	`c3a8d5e1f7b2_steward_integration.py` `rebuild.py` `models.py`	#12, wr#30	`test_gpu_rebuild`, `test_mlflow_rebuild_preinit`	25 pass	
PRAYOG-90	Rebuild attempt state machine and the pre-init boundary	FR-ML-003, FR-STW-001	Done	`c3a8d5e1f7b2_steward_integration.py` `gpu_runs.py` `rebuild.py` …	#12, wr#31	`test_mlflow_rebuild_preinit`, `test_mlflow_lifecycle`	28 pass	
PRAYOG-91	Submit-by-snapshot through the ordinary scheduling path	—	Done	—	#12, #17	`test_rebuild_scheduling_path`, `test_mlflow_queue_order`	10 pass	
PRAYOG-92	Metric comparator with per-metric tolerances	—	Done	—	#12, #18	`test_mlflow_compare`	8 pass	
PRAYOG-93	RAG pipeline with an authorization-bound context window	NFR-SEC-002, FR-STW-015	Done	`setup_steward.py` `engine.py` `0001_initial_schema.py` …	#11, wr#36	`test_steward_context`	5 pass	Steward suite (outside `main`)

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-94	Diagnostic link when a rebuild fails	NFR-SEC-003, FR-STW-003	Done	`cli.py` `links.py` `s3.py`	#11, #26, wr#39	`test_diagnostic_session_audit`, `test_cli_rebuild_reaches_steward`	7 pass, 1 fail	Steward suite
PRAYOG-95	Run-failure diagnosis with cited evidence	FR-GPU-146	Done	`c3a8d5e1f7b2_steward_integration.py` `lifecycle.py` `models.py` …	#11, #26	`tests_steward_check`	17 pass	Steward suite (`steward/tests`, kept outside `main` at the team lead's request)
PRAYOG-96	Two-stage sanitiser for ingestion and outbound search	NFR-SEC-006, FR-STW-011	Done	`ports.py` `sanitiser.py` `search.py` …	#11	`tests_steward_check`	17 pass	Steward suite (`steward/tests`, kept outside `main` at the team lead's request)
PRAYOG-97	Output guard: no fabrication, no command execution	FR-GPU-151, FR-STW-013	Done	`__init__.py` `app.py` `service.py` …	#11	`tests_steward_check`	17 pass	Steward suite (`steward/tests`, kept outside `main` at the team lead's request)
PRAYOG-98	Approve or reject a Steward-proposed corrected job	FR-GPU-149	Done	`proposals.py` `owner.py` `s1.py`	#11, #26	`tests_steward_check`	17 pass	Steward suite (`steward/tests`, kept outside `main` at the team lead's request)
PRAYOG-99	Cross-user fault correlation and hardware warnings	FR-GPU-159	Done	`s2.py`	#11	`tests_steward_check`	17 pass	Steward suite (`steward/tests`, kept outside `main` at the team lead's request)
PRAYOG-100	Optimisation advisory after a successful rebuild	FR-STW-006	Done	`gpu_runs.py` `cli.py` `s4.py` …	#11	`tests_steward_check`	17 pass	Steward suite (`steward/tests`, kept outside `main` at the team lead's request)
PRAYOG-101	Prove per-user isolation of ingestion, retrieval and caching	NFR-SEC-001	Done	`0001_initial_schema.py`	#11	`tests_steward_check`	17 pass	Steward suite (`steward/tests`, kept outside `main` at the team lead's request)
PRAYOG-102	Degrade safely when the agent is unavailable	NFR-AVAIL-001, NFR-GPU-055, FR-STW-010	Done	`steward_bridge.py` `steward_worker.py` `lifecycle.py` …	#11	`tests_steward_check`	17 pass	Steward suite (`steward/tests`, kept outside `main` at the team lead's request)
PRAYOG-103	Unit test suite and coverage gate	NFR-GPU-054	In Progress	`__init__.py` `run.py`	—	`test_concurrency`, `test_job_list_batched`, `test_database_unavailable`	11 pass	Test and Coverage reports
PRAYOG-104	Concurrency harness for the claim	NFR-GPU-001	In Progress	—	—	`scheduler/test_claim`, `scheduler/test_adversarial`, `scheduler/test_passes`, `test_concurrency`	79 pass	Concurrency report; `backend/experiments/`
PRAYOG-105	Property tests for the fairness and allocation invariants	NFR-GPU-009	To Do	`config.py`	—	`scheduler/test_scoring`, `scheduler/test_allocation_properties`, `test_pass_equivalence`	31 pass	property-style equivalence of passes; no Hypothesis suite
PRAYOG-106	Integration and end-to-end suite	—	To Do	—	59db073, 9838d1a, 33d2884	`test_mlflow_e2e`, `test_cli_end_to_end`	3 pass	CI e2e step
PRAYOG-107	Adversarial testing	—	To Do	—	—	`scheduler/test_adversarial`	21 pass	Security report section 7 (37 adversarial cases)

Jira	Story / task	Requirement(s)	Status	Code (files citing the requirement)	GitHub	Automated tests	Result 6 Nov	Other evidence
PRAYOG-108	Security testing and threat model	—	To Do	—	—	`scheduler/test_security`	25 pass	Security report
PRAYOG-109	Performance and load experiment with one optimization	NFR-GPU-014	To Do	—	—	`scheduler/test_adversarial`, `test_job_list_batched`	24 pass	Performance report; `backend/experiments/`
PRAYOG-110	Failure and recovery testing	NFR-GPU-042	To Do	—	—	`scheduler/test_liveness`, `test_database_unavailable`, `test_concurrency`	17 pass	Failure and Recovery report; `backend/experiments/`
PRAYOG-111	Five meaningful defects found, fixed and regression-tested	—	To Do	—	—	`scheduler/test_adversarial`, `scheduler/test_security`, `scheduler/test_liveness`	55 pass	Defect history (13 defects)
PRAYOG-112	AI Engineering Log — architecture activities	—	Done	—	9efaf63	—	—	AI Engineering Log (13)
PRAYOG-113	Deliverable 2 package	—	In Progress	—	04b6cee	—	—	DELIVERABLE_2_PACKAGE.md in the repository
PRAYOG-114	Update the traceability matrix to implementation and tests	—	To Do	—	—	—	—	this matrix (11)
PRAYOG-115	Cost estimation, effort model and Gantt chart	—	To Do	—	—	—	—	Costing (08), Gantt (09)
PRAYOG-116	Deliverable 3 package	—	To Do	—	—	—	—	this D3 package (00–13)
PRAYOG-117	Scenario-driven demonstration rehearsal	—	To Do	—	—	—	—	Sprint 3
PRAYOG-118	Engineering retrospective and final report	—	To Do	—	—	—	—	Project report (12)
PRAYOG-119	Final traceability and Jira dashboard	—	To Do	—	—	—	—	this matrix (11), dashboard (12)